Logic in CS Autumn 2025

Problem Sheet 7

S. Krishna

All the questions in this tutorial are meant to give you a flavour of how an automaton can be manipulated to make automata for related languages.

1. We have already studied some standard closure properties of regular languages, such as complementation, union, intersection, reversal, and concatenation. Now, we turn to two additional important closure properties: **homomorphisms** and **inverse homomorphisms**.

   Definition 1 (Homomorphism) *Let Σ and Γ be alphabets. A function $h : \Sigma \to \Gamma^*$ is called a* homomorphism*. It extends naturally to strings and languages as follows:*

   - *For a string $w = a_1a_2 \cdots a_n \in \Sigma^*$, $h(w) = h(a_1)\, h(a_2) \cdots h(a_n)$.*
   - *For a language $L \subseteq \Sigma^*$, $h(L) = \{\, h(w) \mid w \in L \,\}$.*

   Definition 2 (Inverse Homomorphism) *Given a homomorphism $h : \Sigma \to \Gamma^*$, the* inverse homomorphism *of a language $L \subseteq \Gamma^*$ is defined as*

   $$h^{-1}(L) = \{\, w \in \Sigma^* \mid h(w) \in L \,\}.$$

   Prove that regular languages are closed under both homomorphisms and inverse-homomorphisms.

2. Let L be an arbitrary subset of $\{a\}^*$. Prove that L^* is regular.

3. Prove that if $A \subseteq \Sigma^*$ is a regular language, then the language

   $$\text{Prefix}(A) = \{\, w \in \Sigma^* \mid x = wy \text{ for some } x \in A \text{ and } y \in \Sigma^* \,\}$$

   is also regular. (Note that this technique can be extended to define Suffix(L) and Substring(L))

4. Let L be a regular language. Is the language $L_{\frac{1}{2}}$, the set of first halves of strings in L regular? Formally,

   $$L_{\frac{1}{2}} = \{x \mid \exists y, |x| = |y|, xy \in L\}$$

5. Let L be a regular language. Is the language $\mathsf{SquareRoot}(L)$ defined as $\{w \mid w^2 \in L\}$ regular?

6. Let L be a regular language over the alphabet A. Is the language over A, mid-thirds(L) defined as

   $$\{\, v \mid \exists u, w : |u| = |v| = |w| \text{ and } uvw \in L \,\}$$

   regular?

7. Let L be a regular language. Is the language $\mathsf{Cuberoot}(L)$ defined as $\{w \mid w^3 \in L\}$ regular?

   Note: Try to understand the pattern in Q4,5,6,7 and extending the idea to a general n^{th}-root

8. Let L be a regular language. Prove the regularity of the language $\mathsf{Root}(L)$ defined as

$$\{w \mid \exists n \;\; w^n \in L\}$$

9. For any language L, define $cycle(L) = \{uv \mid vu \in L\}$ as the set of all cyclic shifts of words accepted by L. As an example, if $abcd \in L$, then all its cyclic shifts $abcd, dabc, cdab, bcda$ are also in L. Show that if L is regular, so is $cycle(L)$.

10. **Left and right quotients (residuals).** For languages $L, M \subseteq \Sigma^*$ define the left quotient $L\backslash M = \{\, y \mid \exists x \in L,\; xy \in M \,\}$ and the right quotient $M/L = \{\, x \mid \exists y \in L,\; xy \in M \,\}$. Prove that if M is regular (and L arbitrary) then both $L\backslash M$ and M/L are regular.

11. Let $L, M \subseteq \Sigma^*$ be languages over a nonempty finite alphabet Σ. For any $x, y \in \Sigma^*$, the shuffle of x and y is defined by induction as follows:

$$\varepsilon \otimes y = \{y\}, \quad x \otimes \varepsilon = \{x\}, \quad (ax_0) \otimes (by_0) = a \cdot (x_0 \otimes by_0) \;\cup\; b \cdot (ax_0 \otimes y_0),$$

where $x = ax_0$ and $y = by_0$.

It is extended to languages $L \otimes M$ as follows:

$$L \otimes M = \bigcup_{x \in L,\, y \in M} (x \otimes y).$$

Prove that there exists a DFA for recognising $L \otimes M$ if there exist DFAs for recognising L and M.